$$F_a(t) = F_z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_C(t) = C\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de LaPlace

$$F_a(s) = F_z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(t) = \frac{P_a(s) - P_p(s)}{LS}$$

$$F_C(t) = CSP_p(s)$$

$$F_R(t) = \frac{P_p(s)}{R}$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{LS} = CSP_p(s) + \frac{P_p(s)}{R}$$

Procedimiento algebraico

$$\left(\frac{1}{Z} + \frac{1}{LS}\right)P_a(s) = \left(CS + \frac{1}{R} + \frac{1}{Z} + \frac{1}{LS}\right)P_p(s)$$

$$\frac{LS+Z}{LZS}P_a(s) = \frac{CS^2CLRS^2 + L(Z+R)S + RZ}{LRZS}P_p(s)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{L+Z}{LZS}}{\frac{CLRZS^2 + L(Z+R)S + RZ}{LRZS}}$$

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Resultado

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + Rz}{CLRzs^2 + L(z+R)s + Rz}$$

Error estacionario

(Este error se calcula cuando la entrada $[P_a(s)]$ es un escalon unitario, es decir $P_a(s) = 1/s$:

$$e(s) = \lim_{s\to 0} sP_a(s)\left(1 - \frac{P_p(s)}{P_a(s)}\right) = \lim_{s\to 0} \frac{1}{s}\left(1 - \frac{LRs + Rz}{CLRzs^2 + L(z+R)s + Rz}\right) = 0$$

$$e(t) = 0\,V$$

Estabilidad del sistema en lazo abierto

Para analizar la estabilidad del sistema en lazo abierto se debe calcular los polos

$$CLRzs^2 + L(z+R)s + Rz = 0$$

$$a = CLRz$$

$$b = L(z+R)$$

$$c = Rz$$

$$\lambda_{1,2} = \frac{-L(z+R) \pm \sqrt{L^2(z+R)^2 - 4CLR^2z^2}}{2CLRz}$$

Hipotenso

$z = 0.010$

$L = 0.001$

$R = 1.000$

$C = 3.500$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + Rz}{CLRzs^2 + L(z+R)s + Rz} = \frac{0.001s + 0.01}{3.5\times10^{-5}s^2 + 0.001015s + 0.01}$$

$$\lambda_1 = -14.429 + 8.8051i$$

$$\lambda_2 = -14.429 - 8.8051i$$

Por lo tanto, el sistema presenta una respuesta estable y subamortiguada.

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Normotenso

$Z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 1.500$

$$\frac{Pp(s)}{Pa(s)} = \frac{LRs + Rz}{CLRzs^2 + L(z+R)s + Rz} = \frac{-0.001 + 0.01}{1.5 \times 10^{-3} s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -12.062$
$\lambda_2 = -55.272$

Por lo tanto, el sistema presenta una respuesta estable sobreamortiguada.

Hipertenso

$Z = 0.010$
$L = 0.0001$
$R = 1.000$
$C = 1,000$

$$\frac{Pp(s)}{Pa(s)} = \frac{LRs + Rz}{CLRzs^2 + L(z+R)s + Rz} = \frac{-0.001 + 0.01}{0.00001s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -10.0$
$\lambda_2 = -10.0$

Por lo tanto, el sistema presenta una respuesta estable y criticamente amortiguada.

Modelo de ecuaciones integro-diferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{Pa(t) - Pp(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt = C\frac{dPp(t)}{dt} + \frac{Pp(t)}{R}$$

$$\frac{Pa(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt - C\frac{dPp(t)}{dt} = \frac{Pp(t)}{R} + \frac{Pp(t)}{Z}$$

$$Pp(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt - C\frac{dPp(t)}{dt}\right]$$

$$Pp(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt - C\frac{dPp(t)}{dt}\right]\frac{Rz}{Z+R}$$

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